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PHYSICS

1) A solid cube of edge a is melted and moulded in eight identical small solid cubes and are placed on each other on a straight line with the edge of bottom cube on the same horizontal plane on which big cube was placed, then the vertical shift in centre of mass is:

(1) $\frac{3a}{2}$

(2) 2a

(3) $\frac{5a}{2}$

(4) 3a

2) A ball falling freely from height of 5 m, hits horizontal surface. If $e = \frac{3}{4}$, then the time taken by the ball between initial two successive collisions with the surface will be. ($g = 10 m/s^2$)

(1) 1 s

(2) 1.5 s

(3) 2 s

(4) 3 s

3) A particle is kept at rest at top of a sphere of diameter 30 m. When disturbed slightly it slides down. At what height h from bottom, the particle will leave the sphere :-

(1) 25 m

(2) 30 m

(3) 20 m

(4) 15 m

4) Speed of a particle moving on a circular path depends on distance travelled as $v^2 = 4\square$. Find the magnitude of its tangential acceleration when it has covered 2 complete rotations.

(1) 2 m/s^2

(2) 4 m/s^2

(3) $\frac{1}{2} m/s^2$

(4) Data insufficient

5) A body is revolving with a constant speed along a circle. If its direction of motion is reversed but the speed remains the same, then :-
(a) the centripetal force will not suffer any change in magnitude
(b) the centripetal force will have its direction reversed
(c) the centripetal force will not suffer any change in direction

(d) the centripetal force is doubled

(1) a,b

(2) b,c

(3) c,d

(4) a, c

6) A heavy solid sphere is moving down an inclined plane of $\theta = 60^{\circ}$ of coefficient of friction $\mu_s = 0.45$ then its motion is

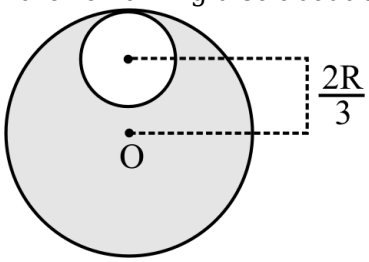
(1) Rolling without slipping

(2) Only slipping

(3) Rolling with slipping

(4) No motion

7) From a circular disc of radius R and mass 9M a small disc of radius $\frac{R}{3}$ is removed as shown in figure. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through O is :



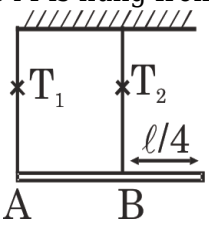
(1) $4\ MR^2$

(2) $\frac{40}{9}MR^2$

(3) $10\ MR^2$

(4) $\frac{37}{9}MR^2$

8) A uniform rod of length and mass M is hung from two strings of equal length from a ceiling as shown.Tension in the two strings is :-

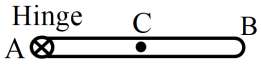


(1) $T_1 = \frac{Mg}{3}$, $T_2 = \frac{2Mg}{3}$

(2) $T_1 = T_2 = \frac{Mg}{3}$

(3) $T_1 = \frac{2Mg}{3}$, $T_2 = \frac{Mg}{3}$

(4) $T_1 = Mg$, $T_2 = 0$

9) 

A uniform rod of mass m and length hinged at end A is released from horizontal position shown in figure. Just after the rod is released :

	Column I		Column II
(A)	Angular acceleration of C	(P)	$\frac{3g}{2}$
(B)	Angular acceleration of B	(Q)	$\frac{3g}{2\ell}$
(C)	Acceleration of C	(R)	$\frac{3g}{4}$
(D)	Acceleration of B	(S)	$\frac{3g}{\ell}$

(1) A → S, B → S, C → R, D → P

(2) A → Q, B → Q, C → R, D → P

(3) A → Q, B → S, C → P, D → R

(4) A → S, B → Q, C → P, D → R

10) A flywheel rotates with a uniform angular acceleration. Its angular velocity increases from 10 π rad/s to 30 π rad/s in 10 seconds. How many rotations did it undergo in this period.

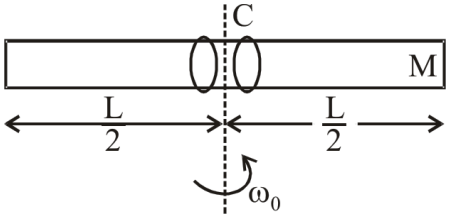
(1) 400

(2) 300

(3) 200

(4) 100

11) A smooth uniform rod of length L and mass M has two identical beads of negligible size, each of mass m, which can slide freely along the rod. Initially, the two beads are at the centre of the rod and the system is rotating with angular velocity ω_0 about an axis perpendicular to rod and passing through the mid-point of rod. There are no external forces. When the beads reach the ends of the rod



the angular velocity of the system is :

(1) $\left(\frac{M}{M+3m}\right)\omega_0$

(2) $\left(\frac{M}{M+6m}\right)\omega_0$

(3) $\left(\frac{M+6m}{M}\right)\omega_0$

(4) ω_0

12) Two concentric shells have masses M and m and their radii are R and r respectively where R > r,

if x be the distance from the common centre, then what is the gravitational potential at a point for which r < x < R :-

(1) $-G\left[\frac{M}{x}+\frac{m}{r}\right]$

(2) $-G\left[\frac{M}{x}-\frac{m}{r}\right]$

(3) $-G\left[\frac{M}{R}+\frac{m}{x}\right]$

(4) $-G\left[\frac{M}{R}-\frac{m}{x}\right]$

13) The rotation of earth having radius R about its axis speeds upto a value such that a man at an angle of 60º from equator feels weightless. The duration of the day in such case will be :-

(1) $8\pi\sqrt{\frac{R}{g}}$

(2) $8\pi\sqrt{\frac{g}{R}}$

(3) $\pi\sqrt{\frac{R}{g}}$

(4) $4\pi\sqrt{\frac{g}{R}}$

14) The orbital angular momentum of a satellite revolving at a distance r from the center of earth is L. If the distance is increased to 16 r, then the new angular momentum will be-

(1) 16 L

(2) 64 L

(3) $\frac{L}{4}$

(4) 4 L

15) The ratio of the velocity required to be given to the satellite at earth's surface to escape earth's gravitational field to the velocity required for the satellite so that the satellite moves in a circular orbit just above earth atmosphere is :-

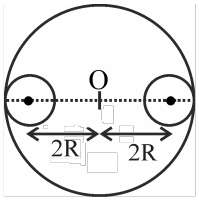
(1) 1

(2) $\sqrt{2}$

(3) $\frac{1}{\sqrt{2}}$

(4) Infinite

16) A solid sphere of uniform density ρ and radius 4 R is placed with its centre at the origin. Two sphere of equal radius R are taken out of the solid sphere leaving behind spherical cavities as shown

in the figure. Intensity due to this object at origin is :- 

(1) 0

(2) $\frac{2}{3}\pi R\rho G$

(3) $\frac{1}{3}\pi R\rho G$

(4) $\frac{1}{4}\pi R\rho G$

17) On dipping a capillary of radius 'r' in water, water rises upto a height H and potential energy of water is u_1. If a capillary of radius 2r is dipped in water, then the potential energy is u_2. The ratio $\frac{u_1}{u_2}$ is

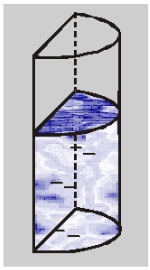
(1) 2 : 1

(2) 1 : 2

(3) 4 : 1

(4) 1 : 1

18) A liquid is contained in a vertical tube of semicircular cross-section as shown in figure. The contact angle is zero. The force of surface tension on the curved part and on the flat part are in ratio



(1) 1 : 1

(2) 1 : 2

(3) π : 2

(4) 2 : π

19) A cube made of material having a density of 0.9×10^3 kg/m^3 floats between water and a liquid of density 0.8×10^3 kg/m^3, which is immiscible with water. What part of the cube is immersed in water ?

(1) $\frac{1}{2}$

(2) $\frac{2}{3}$

(3) $\frac{3}{4}$

(4) $\frac{3}{7}$

20) A tank full of water has a small hole at its bottom. If three-fourth of the tank is emptied in t_1 second and the remaining one-fourth of the tank is emptied in t_2 second. Then, the ratio t_1/t_2 is :-

(1) $\sqrt{3}-1$
(2) 1
(3) $1/\sqrt{2}$
(4) $1/\sqrt{3}$

21) In a test experiment on a model aeroplane in a wind tunnel, the flow speeds on the upper and lower surfaces of the wings are 70 ms^{-1} and 63 ms^{-1} respectively. What is the lift on the wing, If total area of wings is 2.5 m^2 ?
Take the density of air to be 1.3 kgm^{-3} .

(1) 1513 N
(2) 1513 Dynes
(3) 151.3 N
(4) 151.3 Dynes

22) The length of an elastic string is 5 metre when the longitudinal tension is 4 N and 6 metre when the tension is 5 N. If the length of the string (in metre) is "2X" when the longitudinal tension is 9 N (assume Hooke's law is valid) then the value of X will be :

(1) 5
(2) 6
(3) 7
(4) 8

23) A rubber cord has a cross-section area 1 cm^2 and natural length of $\frac{1}{6}$ m. It is stretched by $\frac{1}{6}$ cm to fire a small object of mass 10 gm. If the Young's modulus (Y) is 6×10^8 N/m^2. What is the velocity of the object when released? Assume Hook's law is valid and neglect gravity.

(1) 10 m/s
(2) 60 m/s
(3) $10\sqrt{6}$ m/s
(4) 20 m/s

24) There is a 1 mm thick layer of water between a plate of area 100 cm^2 and another very big plate. The coefficient of viscosity of water is 0.01 poise. Then, the force required to move the smaller plate with a velocity of 10 cm/sec with respect to the larger plate is:-

(1) 10 dyne
(2) 100 dyne
(3) 1000 dyne

(4) 10, 000 dyne

25) Velocity of an object in rectilinear motion is given as function of time by $v = t - t^3$, where v is in m/s and t is in seconds. Find magnitude of average velocity over the time interval from t = 0 s to t = 2 s is

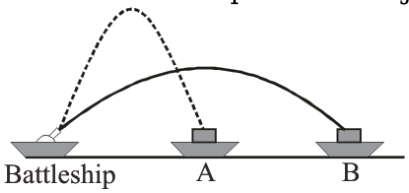
(1) 0 m/s

(2) 1 m/s

(3) 2 m/s

(4) determinable, if initial position is known

26) Two guns on a battleship simultaneously fires two shells with same speed at enemy ships. If the shells follow the parabolic trajectories as shown, which ship will get hit first?



(1) A

(2) B

(3) both at same time

(4) need more information

27) A particle travels such that $\frac{dV}{dt} = -a\sqrt{v}$ where a is constant. Its initial velocity is v^0. The time at which it stops is :-

(1) $\frac{2\sqrt{V_0}}{a}$

(2) $\frac{\sqrt{V_0}}{a}$

(3) $2a\sqrt{V_0}$

(4) $a\sqrt{V_0}$

28) A stone is dropped from the top of a tower and travels 24.5 m in the last second of its journey. The height of the tower is :-

(1) 44.1 m

(2) 49 m

(3) 78.4 m

(4) 72 m

29) The distance between two particles moving towards each other is decreasing at the rate of 6 m/sec. If these particles travel with same previous speed and in the same direction, then the separation increase at the rate of 4 m/sec. The particles have speeds as :-

(1) 5 m/sec ; 1 m/sec

(2) 4 m/sec ; 1 m/sec

(3) 4 m/sec ; 2 m/sec

(4) 5 m/sec ; 2 m/sec

30) A particle moves from the point $\left(2.0\hat{i} + 4.0\hat{j}\right)$ m, at t = 0, with an initial velocity $\left(5.0\hat{i} + 4.0\hat{j}\right)$ ms^{-1}. It is acted upon by a constant force which produces a constant acceleration $\left(4.0\hat{i} + 4.0\hat{j}\right)$ ms^{-2}. What is the distance of the particle from the origin at time 2 s ?

(1) $20\sqrt{2}\,m$

(2) $10\sqrt{2}\,m$

(3) 5 m

(4) 15 m

31) A ball is dropped from a high rise platform at t = 0 starting from rest. After 6 seconds another ball is thrown downwards from the same platform with a speed v. The two balls meet at t = 18s.What is the value of v ? (take g = 10 m/s^2)

(1) 60 m/s

(2) 75 m/s

(3) 55 m/s

(4) 40m/s

32) A temperature difference can generate e.m.f. in some materials. Let S be the e.m.f. produced per unit temperature difference between the ends of a wire, σ the electrical conductivity and κ the thermal conductivity of the material of the wire. Taking M, L, T, I and K as dimensions of mass, length, time, current and temperature, respectively, the dimensional formula of the quantity $Z = \frac{S^2\sigma}{\kappa}$ is :-

(1) $[M^0L^0T^0I^0K^0]$

(2) $[M^0L^0T^0I^0K^{-1}]$

(3) $[M^1L^2T^{-2}I^{-1}K^{-1}]$

(4) $[M^1L^2T^{-4}I^{-1}K^{-1}]$

33) Choose correct option :

(1) No. of significant figure in 2300 is two

(2) Value of (12.5849 + 2.4) is 15.0

(3) No. of significant figure in 7.2500 is five

(4) All of above

34) The pitch of a screw gauge is 0.5 mm and there are 100 divisions on its circular scale. The instruments reads 2nd circular scale division when nothing is put in-between its jaws. In measuring the diameter of a wire, there are 8 divisions on the main scale and 83rd division coincides with the reference line. Find the diameter of the wire (in mm).

(1) 4.415 mm

(2) 4.405 mm

(3) 8.415 mm

(4) 8.405 mm

35) A force acting on a particle varies with the displacement x as $F = ax - bx^2$. Where a = 1 N/m and $b = 1\ N/m^2$. The work done by this force for the first one meter (F is in newtons, x is in meters) is :

(1) $\frac{1}{6}J$

(2) $\frac{2}{6}J$

(3) $\frac{3}{6}J$

(4) $\frac{1}{4}J$

36) A spring with spring constant K when compressed by 1 cm, the potential energy stored is U. If it is further compressed by 3 cm, then change in its potential energy is :

(1) 3U

(2) 9U

(3) 8U

(4) 15U

37) The potential energy of a particle in a force field is $U = \frac{A}{r^3} - \frac{B}{r^2}$, where A and B are positive constants and r is the distance of particle from the centre of field. For the equilibrium, the distance of particle is :-

(1) $\frac{2A}{3B}$

(2) $\frac{3A}{2B}$

(3) $\frac{A}{3B}$

(4) $\frac{A}{2B}$

38) A body is initially at rest. It undergoes one-dimensional motion with constant acceleration. The power delivered to it at time t is proportional to-

(1) $t^{1/2}$

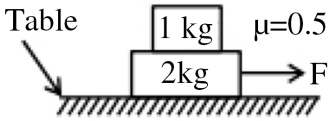
(2) t

(3) $t^{3/2}$

(4) t^2

39) The coefficient of static friction between two blocks is 0.5 and the table is smooth. The maximum

horizontal force that can be applied to move the blocks together is _____N.



(take g = 10 ms^{-2})

(1) 45

(2) 35

(3) 25

(4) 15

40) A small block starts slipping down from a point B on an inclined plane AB, which is making an angle θ with the horizontal, section BC is smooth and the remaining section CA is rough with a coefficient of friction μ. It is found that the block comes to rest as it reaches the bottom (point A) of the inclined plane. If BC = 2AC, the coefficient of friction is given by μ = k tan θ. The value of k is

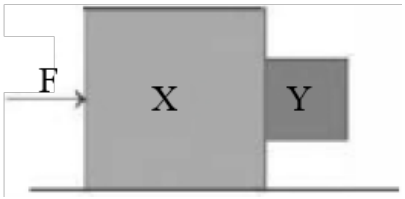
(1) 13

(2) 8

(3) 3

(4) 17

41) Consider the system shown below. A horizontal force F is applied to a block X of mass 8 kg, such that the block Y of mass 2 kg adjacent to it does not slip downwards under gravity. There is no friction between the horizontal plane and the base of the block X. The coefficient of friction between the surfaces of blocks X and Y is 0.5. The minimum value of F is ..... N (take, acceleration due to gravity to be 10 ms^{-2})



(1) 200

(2) 160

(3) 40

(4) 240

42) A heavy box is to dragged along a rough horizontal floor. To do so, person A pushes it at an angle 30° from the horizontal and requires a minimum force F_A, while person B pulls the box at an angle 60° from the horizontal and needs minimum force F_B. If the coefficient of friction between the box and the floor is $\frac{\sqrt{3}}{5}$ the ratio $\frac{F_A}{F_B}$ is

(1) $\sqrt{3}$

(2) $\frac{5}{\sqrt{3}}$

(3) $\sqrt{\frac{3}{2}}$

(4) $\frac{2}{\sqrt{3}}$

43) A small steel ball falls through a syrup at constant speed of 10 cm/s. If the steel ball is pulled upwards with a force equal to twice its effective weight, how fast will it move upwards?

(1) 10 cm/s

(2) 20 cm/s

(3) 5 cm/s

(4) Will not move upward

44) A vernier callipers has 20 divisions on the vernier scale which coincide with 19 divisions on the main scale. The least count of the instrument is 0.2 mm. Then find out the value of main scale division :-

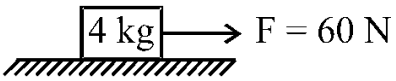
(1) 0.5 mm

(2) 4 mm

(3) 2 mm

(4) $\frac{1}{4}$mm

45) If the acceleration of block is 10 m/sec^2 then coefficient of kinetic friction between block and the surface is :



(1) 0.5

(2) 0.6

(3) 0.4

(4) 0.8

CHEMISTRY

1) **Assertion :** At room temperature O_2 and N_2 are stable molecule but S_2 and P_2 are less stable.
Reason : $2p\pi - 2p\pi$ collateral overlapping is effective but $3p\pi - 3p\pi$ overlapping is weak.

(1) Both assertion and reason are correct and reason is the correct explanation of assertion.

(2) Both assertion and reason are correct but reason is not correct explanation of assertion.

(3) Assertion is correct, reason is incorrect

(4) Both assertion and reason are incorrect.

2) Total number of molecules or ions which have any bond angle of 120°.
I_3^-, BCl_3, PCl_5, CO_3^{-2}, CH_3^-, SF_6, XeF_2, SO_3

(1) 3

(2) 4

(3) 5

(4) 6

3)

Match List-I with List-II correctly :

List I (Compound)		List II (Structure)	
(A)	$XeOF_2$	(P)	Square planar
(B)	PCl_5	(Q)	Tetrahedral
(C)	IF_5	(R)	Trigonal Bipyramidal
(D)	CCl_4	(S)	Square pyramidal
(E)	XeF_4	(T)	T-shaped/Bent 'T'

(1) A-T, B-S, C-R, D-Q, E-P

(2) A-T, B-R, C-S, D-Q, E-P

(3) A-T, B-R, C-S, D-P, E-Q

(4) A-S, B-R, C-T, D-Q, E-P

4) **Assertion:** In PCl_5 axial bonds have been found to be slightly longer than equatorial bonds.
Reason : Axial bond pairs suffer more repulsive interaction from equatorial bond pairs

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

5) Which of the following correct order ?

(1) C=O>C≡O>C=C>C≡N (Bond strength)

(2) $Cl_2 > Br_2 > F_2 > I_2$ (Bond length)

(3) $C–C > \ddot{S}–\ddot{S} > \ddot{O}–\ddot{O}$ (Bond strength)

(4) $C \leftrightharpoons O$ > C=N > F–F > O–O (Bond length)

6) Pick the sets which has correct dipole moment order :-
(A) $CH_3Cl > CHCl_3$ (B) $CH_3Cl > CH_3F$
(C) $H_2O_2 > H_2O$ (D) $H_2O < NH_3 > NF_3$
(E) $SO_2 > SO_3$ (F) $HF > NF_3 > NH_3$

(1) A, B, C, E

(2) A, B, C, F

(3) B,C,E,F

(4) C, D, E, F

7) Which of the following is the correct order of boiling point -

(1) $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$

(2) $PH_3 > NH_3 > AsH_3 > SbH_3 > BiH_3$

(3) $PH_3 > AsH_3 > NH_3 > SbH_3 > BiH_3$

(4) $BiH_3 > SbH_3 > NH_3 > AsH_3 > PH_3$

8) Which of the following statement is wrong if N_2 and O_2 are converted into monoanions N_2^- and O_2^- respectively,

(1) In N_2^-, the N–N bond weakens

(2) In O_2^-, the O–O bond order decrease

(3) In O_2^-, O-O bond length increases

(4) N_2^- becomes diamagnetic

9) If the ionic potential of ions is in the order $M_a^+ > M_b^+ > M_c^+$ and polarizability of ions is in the order $x^- > y^- > z^-$. select the combination of ions having maximum covalent character :

(1) $M_a^+x^-$

(2) $M_b^+x^-$

(3) $M_b^+y^-$

(4) $M_c^+x^-$

10) **Statement-I :-** Formal charge help in the selection of the lowest energy structure from a number of possible Lewis structures for a given specie.
Statement-II :- Generally the lowest energy structure is the one with the smallest formal charge on the atoms.

(1) Both Statement-I and Statement-II are incorrect.

(2) Both Statement-I and Statement-II are correct.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

11) The correct option regarding the following electronic configuration :-

(1) [Rn] $7s^2\ 6d^2$ – p block

(2) [Xe] $6s^2\ 4f^{14}\ 5d^1$ – f block

(3) [Xe] $6s^2\ 4f^{14}\ 5d^{10}\ 6p^1$ – d block

(4) [Xe] $6s^2\ 4f^1\ 5d^1$ – d block

12) Correct order of atomic radii of C, Cs, Al, S is ?

(1) S < C < Al < Cs

(2) S < C < Cs < Al

(3) C < S < Cs < Al

(4) C < S < Al < Cs

13) Electronic configuration of valence shell of four elements A, B, C and D are :-
(A) $ns^2\ np^1$ (B) $ns^2\ np^2$
(C) $ns^2\ np^3$ (D) $ns^2\ np^4$
Correct increasing order of their second ionisation energies is :-

(1) A < B < C < D

(2) A < B < D < C

(3) B < A < D < C

(4) B < A < C < D

14) Among halogens, the correct order of amount of energy released in electron gain (electron gain enthalpy) is :-

(1) F > Cl > Br > I

(2) F < Cl < Br < I

(3) F < Cl > Br > I

(4) F > Cl < Br > I

15) **Statement I :** Acidic strength increases in the order given as HF << HCl << HBr << HI.
Statement II : As the size of the elements F, Cl, Br, I increases down the group, the bond strength of HF, HCl, HBr and HI decreases and so the acid strength increases.
Choose the **correct** answer-

(1) Both **Statement I** and **Statement II** are true.

(2) Both **Statement I** and **Statement II** are false.

(3) **Statement I** is correct but **Statement II** is false.

(4) **Statement I** is incorrect but **Statement II** is true.

16) Match the following :

	Oxide		**Nature**
(A)	CO	(i)	Basic
(B)	BaO	(ii)	Neutral
(C)	Al_2O_3	(iii)	Acidic
(D)	Cl_2O_7	(iv)	Amphoteric

(1) A-ii ; B-i ; C-iv ; D-iii

(2) A-iii ; B-iv ; C-i ; D-ii

(3) A-iv ; B-iii ; C-ii ; D-i

(4) A-i ; B-ii ; C-iii ; D-iv

17) Which of the following is the correct IUPAC name of the recently discovered element of carbon family ?

(1) Unnilquadium

(2) Ununquadium

(3) Ununseptium

(4) Unniloctium

18) Which of the following statements is not correct ?

(1) Number of radial nodes is (n – □ – 1) for any orbital.

(2) Number of angular nodes is (□) for any orbital.

(3) Total number of nodes = (n – 2).

(4) Energy of s–orbitals increases with the increase in the total number of nodes.

19) Arrange the following in increasing order of energy :-
(i) n = 4, □ = 2, m = –1, s = +½
(ii) n = 3, □ = 2, m = –1, s = –½
(iii) n = 4, □ = 0, m = 0, s = +½
(iv) n = 5, □ = 0, m = 0, s = –½

(1) (i) < (ii) < (iii) < (iv)

(2) (iii) < (ii) < (iv) < (i)

(3) (iii) < (iv) < (ii) < (i)

(4) (ii) < (iii) < (i) < (iv)

20) The equilibrium constants for the formation of oxides of nitrogen are given as under (at 25°C):
(a) $N_2 + O_2 \rightleftharpoons 2NO$; $K = 5 \times 10^{-31}$
(b) $N_2 + 2O_2 \rightleftharpoons 2NO_2$; $K = 8 \times 10^{-31}$
(c) $2N_2 + 5O_2 \rightleftharpoons 2N_2O_5$; $K = 1 \times 10^{-34}$
Choose the correct statement :-

(1) NO_2 is the least stable oxide

(2) N_2O_5 is the most stable oxide

(3) The stability order should be : $N_2O_5 > NO_2 > NO$

(4) The stability order should be : $NO_2 > NO > N_2O_5$

21) If $A \rightleftharpoons B$; $K_1 = 100$, $3C \rightleftharpoons 3B$; $K_2 = 10^6$, $\frac{1}{2}C \rightleftharpoons \frac{1}{2}D$; $K_3 = 10^{-4}$, then find out K for $A \rightleftharpoons D$:-

(1) 10^2

(2) 10^{-8}

(3) 10^8

(4) 10^6

22)

The degree of dissociation of PCl_5 (α) obeying the equilibrium, $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, is approximately related to the pressure at equilibrium by (given $\alpha << 1$) :

(1) $\alpha \propto P$

(2) $\alpha \propto \frac{1}{\sqrt{P}}$

(3) $\alpha \propto \frac{1}{P^2}$

(4) $\alpha \propto \frac{1}{P^4}$

23) $Br_2 \rightarrow Br^- + BrO_3^-$

Equivalent weight of Br_2 is :-

(1) $\frac{M}{2}$

(2) $\frac{M}{10}$

(3) $\frac{3M}{5}$

(4) $\frac{M}{8}$

24)

Calculate the correct increasing order of average oxidation number of sulphur in :
S_8,$S_2O_3^{2-}$, $S_2O_7^{-2}$, $S_4O_6^{-2}$, SO_3^{-2} :-

(1) $S_8 < S_2O_3^{-2} < S_4O_6^{-2} < SO_3^{-2} < S_2O_7^{-2}$

(2) $S_2O_3^{-2} < S_4O_6^{-2} < S_8 < SO_3^{-2} < S_2O_7^{-2}$

(3) $S_4O_6^{-2} < S_2O_3^{-2} < S_8 < SO_3^{-2} < S_2O_7^{-2}$

(4) $S_8 < S_4O_6^{-2} < S_2O_3^{-2} < SO_3^{-2} < S_2O_7^{-2}$

25) Oxidation state of oxygen in CrO_5 is :-

(1) –1

(2) –2

(3) Both (1) & (2)

(4) $-\frac{1}{2}$

26) How many mole of $K_2Cr_2O_7$ are required to oxidise 12 mole of SnC_2O_4 in acidic medium.

(1) 4

(2) 2

(3) 6

(4) 8

27) If wavelength of first lyman line for H-atom is 'x' then determine wavelength of limiting line of Balmer series :-

(1) $\frac{x}{3}$

(2) 3x

(3) $\frac{27x}{5}$

(4) $\frac{5x}{27}$

28) Uncertainty in the position of an electron (mass = 9.1×10^{-31} Kg) moving with a velocity 300 ms^{-1}, having an uncertainity in velocity of 0.001%, will be :- (h = 6.63×10^{-34} Js)

(1) 5.76×10^{-2} m

(2) 1.92×10^{-2} m

(3) 3.84×10^{-2} m

(4) 19.2×10^{-2} m

29) A power source of 360 watt emits 50% of energy as light (λ = 331 nm). If photons are emitted out by the source per second is $x \times 10^{20}$ then what is the value of x ? (Given : h = 6.62×10^{-34} Js, c = 3×10^{8} m/s)

(1) 30

(2) 60

(3) 3

(4) 6

30) Which of the following has maximum mass :-

(1) 0.1 g atom of C

(2) 0.1 mole of NH_3

(3) 6.02×10^{22} molecule of H_2 gas

(4) 1120 ml of CO_2 at 1 atm, 273K

31) 2 mol of H_2S and 11.2 L of SO_2 at STP react to form X mole of sulphur according to given reaction

$SO_2 + 2H_2S \rightarrow 3S + 2H_2O$

The value of X is :

(1) 1.5

(2) 3

(3) 11.2

(4) 6

32) 100 kg of iron ore (Fe_2O_3), containing 20% impurities, on reduction with CO give iron equal to :- [At. wt. of Fe = 56]

$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

(1) 112 kg
(2) 80 kg
(3) 160 kg
(4) 56 kg

33) An organic compound contains 4% sulphur. Its minimum molecular weight is :

(1) 200
(2) 400
(3) 800
(4) 1600

34) Which of the following is correct option for free expansion of an ideal gas under adiabatic condition?

(1) $q = 0, \Delta T \neq 0, w = 0$
(2) $q \neq 0, \Delta T = 0, w = 0$
(3) $q = 0, \Delta T = 0, w = 0$
(4) $q = 0, \Delta T < 0, w \neq 0$

35) $\Delta S_{Total} = -40 \text{ kJ mol}^{-1} \times K^{-1}$
$\Delta H_{sys} = 2000 \text{ kJ mol}^{-1}$
T = 400 K
Calculate value of ΔS_{system} ?

(1) $-35 \text{ kJ mol}^{-1} K^{-1}$
(2) $-5 \text{ kJ mol}^{-1} K^{-1}$
(3) $-40 \text{ kJ mol}^{-1} K^{-1}$
(4) $+ 5 \text{ kJ mol}^{-1} K^{-1}$

36) A reaction is spontaneous if :-

(1) $T\Delta S < \Delta H$ and both ΔH and ΔS are +ve
(2) $T\Delta S > \Delta H$ and ΔH is +ve and ΔS is –ve
(3) $T\Delta S > \Delta H$ and both ΔH and ΔS are +ve
(4) $T\Delta S = \Delta H$ and both ΔH and ΔS are +ve

37) Atomisation enthalpy of CH_4 is 360 kJ/mol and C_2H_6 has 620 kJ/mol. Then bond dissociation energy of C–C bond is :-

(1) 170 kJ/mol
(2) 50 kJ/mol
(3) 80 kJ/mol
(4) 220 kJ/mol

38) The heat of combustion of CH_4 is –400 kJ mol^{-1}. Calculate the heat released when 40 g of H_2O is formed upon combustion :

(1) 222.2 kJ

(2) 100 kJ

(3) 444.4 kJ

(4) 888.8 kJ

39) The value of ΔH & ΔS for the reaction :-
$C_{graphite} + CO_{2(g)} \rightarrow 2CO_{(g)}$
are 170 KJ & 170 J K^{-1} respectively.This reaction will be spontaneous at

(1) 510 K

(2) 710 K

(3) 910 K

(4) 1110 K

40) Calculate the solubility of silver phosphate (Ag_3PO_4) in 0.1 M $AgNO_3$
(K_{sp} of Ag_3PO_4 = 1.1×10^{-16}) :-

(1) 1×10^{-8} M

(2) 4×10^{-15} M

(3) 1.1×10^{-15} M

(4) 1.1×10^{-13} M

41) In which of the following triad properties are intensive:-

(1) Pressure, temperature, mass

(2) Density, volume, internal energy

(3) Temperature, density, pressure

(4) Pressure, volume, enthalpy

42) Calculate the $[OH^-]$ in 0.01M aqueous solution of NaOCN (K_b for OCN^- = 10^{-10}) :

(1) 10^{-6} M

(2) 10^{-7} M

(3) 10^{-8} M

(4) None of these

43) pH of the resulting solution obtained by mixing equal volumes of two solutions with pH = 3 and pH = 6 is

(1) 4.5

(2) 3.0

(3) 9.0

(4) 3.3

44) The dissociation of phosgene, which occurs according to the reaction
$COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$
is an endothermic process. Which of the following will increase the degree of dissociation of $COCl_2$?

(1) Adding Cl_2 to the system

(2) Adding helium to system at constant volume

(3) Decreasing the temperature of the system

(4) Reducing the total pressure

45) $X_2O_{4(\square)} \rightarrow 2XO_{2(g)}$, $\Delta E = 2.1$ kcal, $\Delta S = 20$ cal k^{-1} at 300 K what is the value of ΔG :-

(1) 2.7 kcal

(2) – 2.7 kcal

(3) 9.3 kcal

(4) – 9.3 kcal

BIOLOGY

1) (A) All eukaryotic cells are not identical
(B) Cells differ greatly in size, shape and activities.
(C) The shape of the cell may vary with the function they perform
(D) The shape, size and number of mitochondria is variable depending upon physiological activity of the cells.
(E) Smaller and less nucleoli are present in cells actively carrying out protein synthesis.
Identify the correct statement

(1) Only A and C

(2) Only A,B and D

(3) Only B,C and D

(4) Except E, all are correct

2) Microtubule, microfilaments and intermediate filaments constitute-

(1) Elaborate network of filamentous proteinaceous structure in cytoplasm.

(2) Dense granular structure in cytoplasm.

(3) Network of tiny tubular membranous structures scattered in cytoplasm

(4) Extensive compartmentalisation of cytoplasm due to presence of membrane bound organelle.

3) Consider the following four statement (a - d) and select the option which includes all the correct ones :
(a) Many membrane bound minute vesicles called microbodies that contain various enzymes, are present in both plant and animal cells.
(b) Cilia and flagella are hair-like out growths of the cell membrane.
(c) The stroma of the chloroplast contains enzymes required for the synthesis of carbohydrates and proteins.
(d) Anton von Leenwenhoek first saw and described a dead cell.

Options :

(1) Statement b, c and d only
(2) Statement a, b only
(3) Statement a, b and c only
(4) Statement c, d only

4) From the following select the statement that is true :-

(1) All cells have a cell wall
(2) Animal cells contain microtubules but plant cells do not contain microtubules
(3) The golgi apparatus is found only in animal cells
(4) Chloroplasts are found in plant cells but not in prokaryotic and animal cells

5) Match the Column-I with Column-II and find the correct answer.

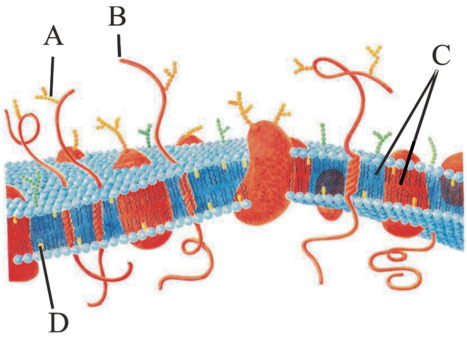
	Column-I		**Column-II**
(i)	Mycoplasma	(a)	0.02-0.2 µm
(ii)	Bacteria	(b)	0.3 µm
(iii)	Red blood cell	(c)	3-5 µm
(iv)	Virus	(d)	7 µm

(1) (i)-d, (ii)-c, (iii)-a, (iv)-b
(2) (i)-b, (ii)-c, (iii)-d, (iv)-a
(3) (i)-b, (ii)-d, (iii)-c, (iv)-d
(4) (i)-a, (ii)-b, (iii)-c, (iv)-d

6) Two statements are given below :
Statement-1 :- The inner and outer nuclear membranes remains continous with rough endoplasmic reticulum.
Statement-2 :- The nucleus is enclosed by nuclear envelops, a double membrane structure with nuclear pores.
Options :

(1) Statement-1 is true while statement-2 is false.
(2) Statement-2 is true while statement-1 is false.
(3) Both Statement 1 and 2 are true.
(4) Both Statement 1 and 2 are false.

7) Identify the A, B, C and D select the correct statement regarding A, B, C and D :-



(1) Percentage of "A" is maximum in the membrane

(2) Percentage of "B" is minimum in the membrane

(3) Lateral movement of "A" help in the measurement of fluidity of membrane

(4) The fluidity of membrane is due to "C"

8) Match the column-I and II and choose the correct option :-

	Column-I		**Column-II**
(A)	Neutral solutes	(i)	Active transport
(B)	Water	(ii)	Facilitated diffusion
(C)	Polar molecules	(iii)	Osmosis
(D)	Na^+ / K^+ pump	(iv)	Simple diffusion

(1) (A) - i, (B) - ii, (C) - iii, (D) - iv

(2) (A) - iv, (B) - i, (C) - iii, (D) - ii

(3) (A) - iv, (B) - iii, (C) - ii, (D) - i

(4) (A) - iii, (B) - ii, (C) - i, (D) - iv

9) Match the following columns and find out correctly matched option :-

	Column-I		**Column-II**
(a)	G_0 phase	(i)	Cells metabolically active but don't proliferate, also called quiscent phase.
(b)	G_1 phase	(ii)	Synthesis of cellular component and it is also an interval between mitosis and DNA replication.
(c)	G_2 phase	(iii)	Synthesis of tubulin protein for mitosis and cell growth continues.
(d)	S-phase	(iv)	Synthesis or replication of DNA, amount of DNA doubles per cell.

(1) a-(i), b-(ii), c-(iii), d-(iv)

(2) a-(iv), b-(iii), c-(ii), d-(i)

(3) a-(i), b-(iv), c-(iii), d-(ii)

(4) a-(ii), b-(iv), c-(i), d-(iii)

10) Match the column-I with column-II -

	Column-I		**Column-II**
i	Leptotene	(a)	Terminalisation of chiasmata
ii.	Zygotene	(b)	First stage of prophase-I
iii.	Diakinesis	(c)	Crossing-over
iv.	Pachytene	(d)	Bi-valent / tetrad

(1) i–b, ii–d, iii–c, iv–a

(2) i–b, ii–c, iii–d, iv–a

(3) i–b, ii–d, iii–a, iv–c

(4) i–b, ii–a, iii–c, iv–d

11) Match the following columns.

	Column-I		**Column-II**
A.	Disintegration of nuclear membrane	P.	Anaphase
B.	Appearance of nucleolus	Q.	Prophase
C.	Division of centromere	R.	Telophase
D.	Chromosome duplication	S.	S-phase

Codes :

	A	**B**	**C**	**D**
(1)	Q	R	P	S
(2)	Q	R	S	P
(3)	R	Q	P	S
(4)	R	Q	S	P

(1) 1

(2) 2

(3) 3

(4) 4

12)

(a) Golgi complex is related to cell secretions.
(b) Lysosome is the part of endomembrane system.
(c) Mitochondria is associated with light reaction.
(d) Ribosome synthesize RNA.

In given statements which of the following are incorrect :-

(1) a and d

(2) b and d

(3) b and c

(4) c and d

13) Given below are two statements -
Statement-I :- Although in many cases, in telophase-I the chromosomes do undergo some dispersion but they do not reach the extremely extended state of interphase nucleus.
Statement-II :- Meiosis decreases the genetic variability in the population of organism from one generation to the next.
In the light of the above statements, choose the correct answer from the options given below -

(1) Both statement I and statement II are correct

(2) Both statement I & statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

14) What is the correct sequence of the steps given here ? Also identify the process depicted in the steps.
i. Homologous chromosomes move toward opposite poles of the cell
ii. Chromosomes gather at the two poles of the cell and the nuclear membranes reform
iii. Homologous chromosomes form pairs and exchange segments
iv. Homologous chromosomes align on equatorial plate

(1) The correct sequence is iii → iv → i → ii and the process is meiosis-I

(2) The correct sequence is ii → i → iv → iii and the process is mitosis

(3) The correct sequence is iv → i → iii → ii and the process is meiosis-I

(4) The correct sequence is iii → iv → i → ii and the process is meiosis-II

15) How many chromosomes will be found in a cell at anaphase II if cell having 2n = 40 ?

(1) 40

(2) 80

(3) 160

(4) 320

16) **Assertion :** Fatty acids could be unsaturated without double bonds or saturated with one or more C = C double bonds.
Reason : Fatty acids are found esterified with glycerol which is trihydroxy methane.

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true but reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false

(4) Both assertion and reason are false.

17) **Assertion :** Inhibition of succinic dehydrogenase by malonate is an example of competitive inhibition.
Reason : When the inhibitor closely resembles the substrate in its molecular structure and inhibit the activity of enzyme, it is known as competitive inhibitor.

(1) Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Assertion & Reason are False.

18) Find out the correct match from the following table :-

	Column I	**Column II**	**Column III**
(i)	α helix	Keratin	Intra molecular hydrogen bond
(ii)	β pleated sheet	Fibroin	Inter molecular hydrogen bonding
(iii)	Haemo globin	Tertiary structure	2 Polypeptide chain

(1) i only
(2) i and ii only
(3) i, ii and iii
(4) iii only

19) Match the following with respect to the average composition of cells

Component		**% of the total cellular mass**	
a	Proteins	i	5-7
b	Carbohydrates	ii	3
c	Lipids	iii	10-15
d	Nucleic acids	iv	2

(1) a(i), b(iii), c(ii), d(iv)
(2) a(iii), b(ii), c(iv), d(i)
(3) a(iii), b(i), c(iv), d(ii)
(4) a(i), b(ii), c(iii), d(iv)

20) Read the following carefully :
(A) Enzymes are denatured at high temperature but in certain exceptional organisms they are effective even at temperature 80–90°C.
(B) Enzymes are highly specific.
(C) Most enzymes are proteins but some are lipids
(D) Enzymes require optimum pH for maximum activity.
How many statement is/are wrong ?

(1) 4
(2) 3

(3) 2

(4) 1

21) Match the column-I & II and choose the correct option ?

	Column-I		Column-II
1.	Heteropolysaccharide	a	Cellulose
2.	Homopolysaccharide	b	Hyaluronic acid
3.	Sulphur containing amino acid	c	Rubber
4.	Polymeric substance	d	Maltose
5.	Reducing sugar	e	Methionine

(1) 1→a, 2→b, 3→c, 4→d, 5→e

(2) 1→b, 2→a, 3→e, 4→c, 5→d

(3) 1→a, 2→b, 3→e, 4→c, 5→d

(4) 1→b, 2→a, 3→c, 4→d, 5→e

22) Which of the following is true for competitive inhibition of enzymes :

(1) Km increases and V_{max} remains unchanged

(2) Km decreases and V_{max} remain unchanged

(3) Km remains unchanged and V_{max} decreases

(4) Km remains unchanged and v_{max} increases

23) Select incorrect statement :-

(1) 'Animals', 'Mammals' and 'dogs' represent taxa at different levels.

(2) Carnivora, includes families like Felidae and Canidae

(3) As we go from species to kingdom number of common characters increaces

(4) Wheat belongs to order poales

24) (A) Properties of tissues are present in constituent cell & do not arise as a result of interactions among constituent cells.
(B) Properties of cellular organelles are not present in the molecular constituents of the organalles but arise as a result of interaction among the molecular components comprising the organelle

(1) A – Correct ; B – Correct

(2) A – Incorrect ; B – Correct

(3) A – Correct ; B – Incorrect

(4) A – Incorrect ; B – Incorrect

25) Which one of the following statement is incorrect ?

(1) *Panthera* genus is similar to another genus canidae which includes cats.

(2) Lion (*Panthera leo*), leopard (*P.pardus*) and tiger (*P.tigeris*) with several common characters are all species of the genus *panthera*.

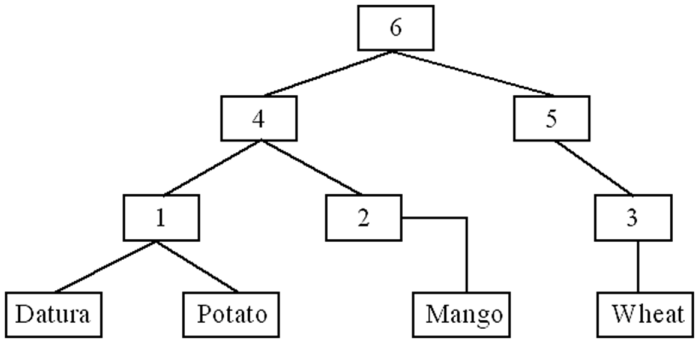
(3) Genera are aggregation of closely related species.

(4) Each genus may have one or more than one specific epithets representing different organisms, but having morphological similarities.

26) Read the following statements :
A. For plants, scientific names are based on agreed principles and criteria, which are provided in the ICBN.
B. Each rank or taxon, in fact, represents a unit of classification in taxonomical hierarchy.
C. Genera are aggregates of closely related families.
D. Lower the category, greater is the difficulty of determining the relationship to other taxa at the same level.
How many of the above statements are **not** correct :-

(1) Four
(2) Three
(3) Two
(4) One

27) Find the correct option in below diagram according to taxonomical hierarchy.



(1) 1-*Solanum*, 2-Poales, 3-Sapindales, 4- Monocotyledonae, 5- Dicotyledonae, 6- Angiospermae
(2) 1- Polymoniales, 2-Sapindales, 3-Poales, 4- Dicotyledonae, 5- Monocotyledonae, 6- Angiospermae
(3) 1- Angiospermae, 2-Dicotyledonae, 3- Monocotyledonae, 4- Angiospermae, 5- Poales, 6- Sapindales
(4) 1- *Solanum*, 2-*Mangifera*, 3- *Triticum*, 4- Dicotyledonae, 5- Monocotyledonae, 6- Plantae

28) According to five kingdom classification multicellular eukaryotes are mainly classified on the basis of ?

(1) Phylogeny
(2) Mode of nutrition
(3) Mode of reproduction
(4) Body organisation

29) Which bacteria oxidise various inorganic substances and use the released energy for their ATP

production and they play a great role in recycling of nutrients like nitrogen, phosphorus, iron and sulphur ?

(1) Archaebacteria

(2) Photosynthetic autotroph bacteria

(3) Chemosynthetic autotroph bacteria

(4) All bacteria

30) Some characters are given below. Which of them are true for the cyanobacteria ?
(a) Oxygenic photosynthesis
(b) Non-oxygenic photosynthesis
(c) Oxygenic Chemotropism
(d) Ability of N_2 fixation

(1) a and b

(2) a and c

(3) a and d

(4) b and c

31) An organism having DNA and RNA in its cytoplasm but no cell wall, is :-

(1) Virus

(2) Bacteria

(3) Cyanobacteria

(4) Mycoplasma

32) Archaebacteria tolerate extreme environmental condition due to their special -

(1) Flagella

(2) Cell membrane

(3) Nucleoid

(4) Ribosome

33) Most of them have two flagella one lies longitudinally and the other transversely in a furrow between the wall plates belongs to -

(1) Euglenoids

(2) Protozoans

(3) Dinoflagellates

(4) Diatoms

34) *Trypanosoma* are classified under which of the following kingdom ?

(1) Monera

(2) Protista

(3) Algae

(4) Fungi

35) Match the column-I with column-II and select the correct option :-

Column-I		Column-II	
(A)	Monera	(I)	*Spirogyra, Solanum*
(B)	Protista	(II)	*Bacillus, Nostoc*
(C)	Plantae	(III)	*Euglena, Gonyaulax*
(D)	Fungi	(IV)	*Mucor, Rhizopus*

(1) A-III, B-II, C-I, D-IV

(2) A-II, B-III, C-I, D-IV

(3) A-II, B-IV, C-I, D-III

(4) A-II, B-I, C-IV, D-III

36) Read the following statement (A-D).
(A) Cell wall of Fungi consists of chitin.
(B) Most fungi are heterotrophic
(C) Fungi can also live as symbionts in association with algae as lichens and with roots of higher plants as mycorrhiza
(D) Fusion of two nuclei is called plasmogamy
How many of the above statement are correct ?

(1) Two

(2) Three

(3) Four

(4) One

37) In oogamy, fertilization involves :-

(1) A large motile female gamete and a small non–motile male gamete

(2) A small non–motile female gamete and a large motile male gamete

(3) A large non–molile female gamete and a small motile male gamete

(4) A large non–motile female gamete and a small non–motile male gamete

38) What is the ploidy of rhizoid cells if capsule cells of moss are diploid ?

(1) n

(2) 2n

(3) 3n

(4) 4n

39) Which of the following is vascular cryptogam?

(1) *Cedrus*

(2) *Equisetum*

(3) *Ginkgo*

(4) *Marchantia*

40) In gymnosperms, the endosperm is formed by the :-

(1) Fusion of one male gamete with two polar nuclei

(2) Fusion of one male gamete with one polar nuclei

(3) Fusion of male gametes with the egg

(4) Development of a megaspore

41) Which of the following statements are correct about gymnosperms ?
(A) The gymnosperms are plants in which the ovules are not enclosed by any ovary wall.
(B) Herbs, shrubs and trees are found in gymnosperms.
(C) One of the gymnosperms, the giant redwood red tree *Sequoia* is one of the tallest tree species.
(D) The roots are generally adventitious in gymnosperms.

(1) C and D only

(2) B and D only

(3) A and C only

(4) A and B only

42) Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R) :
Assertion (A) : All gymnosperms are heterosporous.
Reason (R) : The leaves in gymnosperms are well-adapted to withstand extremes of temperature, humidity and wind.
In the light of above statements, choose the correct answer from the options given below :

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(2) (A) is true but (R) is false.

(3) (A) is false but (R) is true.

(4) Both (A) and (R) are true and (R) is the correct explanation of (A).

43) Given below are two statements :
Statement-I : Unlike the gymnosperms where the ovules are naked, in the *Salvia* & other flowering plants the pollen grains and ovules are developed in specialised structure.
Statement-II : *Ginkgo* belongs to gymnosperm & it produce haploid microspores & megaspores.
In the light of the above statements, choose the most appropriate answer from the options given below :-

(1) Both Statement-I and Statement-II are correct.

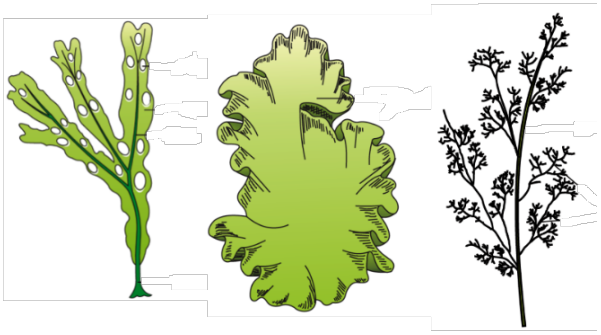
(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

44)

Select the incorrect information given about the algae shown in the given diagrams :-



(1) All of them have cellulose in their cell wall.

(2) Two of them are having floridean starch as stored food material.

(3) At least one of them is having chlorophyll b.

(4) Two of them are having phycoerythrin as photosynthetic pigment.

45) In which one of the following, the genus name, its character and its class are not correctly matched, whereas the remaining three are correctly matched ?

	Genus Name	**Character**	**Class**
(1)	*Trichoderma*	Sexual stage is absent	Deuteromycetes
(2)	*Claviceps*	Endogenous ascospores	Ascomycetes
(3)	*Ustilago*	Septate mycelium	Basidiomycetes
(4)	*Puccinia*	Aseptate and coenocytic mycelium	Phycomycetes

(1) 1

(2) 2

(3) 3

(4) 4

46) Match column-I with column-II and choose the correct option-

	Column-I		**Column-II**
(a)	Air bladder present	(i)	*Bangarus*
(b)	Mammary glands	(ii)	*Ornithorhynchus*
(c)	Neck absent	(iii)	*Bufo*
(d)	Skin cast	(iv)	*Betta*

(1) (a)-i, (b)-ii, (c)-iii, (d)-iv

(2) (a)-iv, (b)-ii, (c)-iii, (d)-i

(3) (a)-iv, (b)-iii, (c)-ii, (d)-i

(4) (a)-i, (b)-iii, (c)-ii, (d)-iv

47) Given below are two statements :-
Statement-I :- Due to absence of air bladder, cartilagenous fishes have to swim constantly to avoid sinking.
Statement-II :- In cartilagenous fishes notochord is persistent throughout life.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both Statement-I and Statement-II are incorrect
(2) Statement-I is incorrect but Statement-II is correct
(3) Statement-I is correct but Statement-II is incorrect
(4) Both Statement-I and Statement-II are correct

48) Which of the following is incorrect match ?

(1) *Torpedo* - electric organs
(2) *Trygon* - Poison sting
(3) *Labeo* - Claspers present
(4) *Pterophyllum* - Aquarium fish

49) In which fish, airbladder regulates buoyancy ?

(1) *Pristis*
(2) *Torpedo*
(3) *Exocoetus*
(4) *Trygon*

50) An examples of vertebrate animals without jaws are :-

(1) *Petromyzon* and *Myxine*
(2) *Petromyzon* and *Scoliodon*
(3) *Scoliodon* and *Myxine*
(4) *Scoliodon* and *Pristis*

51) How many of the following animals are urochordates ?
Ascidia, Brachiostoma, Amphioxus, Doliolum, Lancelet, Salpa

(1) Two
(2) Three
(3) Four
(4) Five

52) Stinging capsules (nematocysts) are found in :

(1) Wasp and physalia
(2) Adamsia and scorpion

(3) Sea pen and sea fan

(4) Meandrina and tapeworm

53) Which one of the following is an exclusive character of class-mammalia ?

(1) Homeothermy

(2) Internal fertilisation

(3) The presence of a four chambered heart

(4) The presence of mammary glands

54) Read column-I, II & III and identify their correct match :-

	Column-I		**Column-II**		**Column-III**
A	Mollusca	P	Spiny bodied animals	i	Balanoglossus, Saccoglossus
B	Echinodermata	Q	Worm like marine animals	ii	Aplysia, Sepia
C	Platyhelminthes	R	Soft bodied animals	iii	Fasciola, Taenia
D	Hemichordata	S	Dosro-ventrally flattened body	iv	Asterias, Antedon

(1) A-R-ii, B-P-iv, C-S-iii, D-Q-i

(2) A-R-iv, B-P-ii, C-S-iii, D-Q-i

(3) A-P-ii, B-R-iv, C-S-iii, D-Q-i

(4) A-R-iv, B-P-iii, C-S-ii, D-Q-i

55) Read the following terminology of column-I & II and identify their correct match :-

Column-I		**Column-II**	
(A)	Dorsal hollow nerve cord	(i)	Echinodermata adult
(B)	Stomochord	(ii)	Porifera mostly
(C)	Radial symmetry	(iii)	Platyhelminthes
(D)	Feather like gills	(iv)	Annelida
		(v)	Chordata
		(vi)	Mollusca
		(vii)	Hemichordata

(1) (A)-(v), (B)-(vii), (C)-(ii), (D)-(iv)

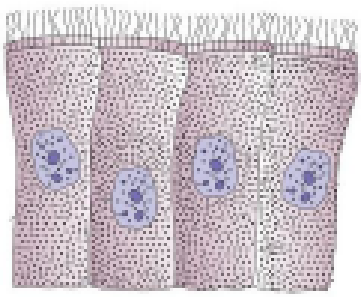
(2) (A)-(v), (B)-(vii), (C)-(vi), (D)-(ii)

(3) (A)-(v), (B)-(vii), (C)-(i), (D)-(vi)

(4) (A)-(vii), (B)-(v), (C)-(iii), (D)-(i)

56) Well marked bioluminescence is found in animals having comb plates. This statement is incorrect for how many animals given below :-
[*Ctenoplana*, Aurelia, Obelia, Physalia, *Pleurobrachia*, Ancylostoma, Ascaris, Enterobius, Tapeworm]

(1) Four
(2) Two
(3) Seven
(4) Six



57) Which of the following is the function of the epithelial tissue shown ?

(1) Help in gaseous exchange
(2) Move particles or mucus in any direction
(3) Move particles or mucus in a specific direction
(4) More than one option are correct

58)

Only tissue which originated from all these germinal layers' ectoderm, mesoderm and endoderm during embryonic development is:-

(1) Connective tissue
(2) Epithelium tissue
(3) Muscular tissue
(4) Neural tissue

59) **Assertion :-** Bones have a hard and non pliable ground substance rich in calcium salts and collagen fibres.
Reason :- the bone cells (osteocytes) are present in the space called lacunae.

(1) Both the assertion and the reason are true and reason are correct explanation of the assertion.
(2) Both the assertion and the reason are true but the reason is not the correct explanation of assertion.
(3) Assertion is correct but reason is not correct.
(4) Assertion is not correct but reason is correct.

60) Which of the following part of alimentary canal of cockroach is responsible for storing of food ?

(1) Gizzard
(2) Oesophagus
(3) Oral-cavity

(4) Crop

61) Genital pouch of male cockroach is bounded dorsally by ___A___ and ventrally by ___B___. Select the correct option for A and B respectively.

(1) 9th and 10th terga, 7th sternum
(2) 7th and 8th terga, 9th sternum
(3) 9th and 10th terga, 9th sternum
(4) 9th and 10th terga, 8th sternum

62) Find out incorrect statement about Cockroach.

(1) Three ganglia lies in thorax, six in abdomen.
(2) Exchange of gases takes place at tracheoles.
(3) Mosaic vision with more resolution but less sensitivity.
(4) Supra-oesophageal ganglion supplies nerves to antennae & compound eye

63) Read the statements with regard to frog. Which of the statement is/are correct and incorrect ?
(i) The medulla oblongata passes out through temporal foramen and continues upto spinal cord
(ii) Vasa efferentia are 10-12 in number that arise from testis.
(iii) Ovaries have no functional connection with kidneys.
(iv) Frogs are uricotelic.

(1) Statement (i), (ii) and (iii) are correct while statement (iv) is incorrect.
(2) Statement (i) and (ii) are correct while statement (iii) and (iv) are incorrect.
(3) Statement (ii) and (iii) are correct while statement (i) and (iv) are incorrect
(4) Statement (ii), (iii) and (iv) are correct while statement (i) is incorrect.

64) Which of the following statement is correct for Rana tigrina ?

(1) Skin is smooth, slippery and dry.
(2) Eye are bulged and covered by nictitating membrane.
(3) Fore limbs are longer than hind limbs
(4) Hind limbs have four digits and fore limbs have five digits.

65) The position of thin walled urinary bladder of frog is ?

(1) Dorsal to the rectum
(2) Ventral to the rectum
(3) Anterior to the rectum
(4) Posterior to the rectum

66) The triangular structure of frog which joins the right atrium is :-

(1) Sinus Venosus
(2) Conus Arteriosus

(3) Ventricle

(4) Left Auricle

67) Match List-I with List-II :-

	List-I		**List-II**
(A)	Eosinophils	(I)	0.5-1%
(B)	Monocytc	(II)	20-25%
(C)	Basophils	(III)	6-8%
(D)	Lymphocytes	(IV)	2-3%

Choose the correct answer from the options given below :-

(1) A-IV, B-I, C-III, D-II

(2) A-I, B-IV, C-III, D-II

(3) A-IV, B-III, C-I, D-II

(4) A-III, B-I, C-IV, D-II

68) **Assertion :** During ventricular systole the tricuspid and bicuspid valves become closed.
Reason : Due to attempted backflow of blood from ventricles into the atria.

(1) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.

(2) Assertion is correct but Reason is incorrect.

(3) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

(4) Assertion is incorrect but Reason is correct.

69) From the following organs how many organs in human receive both oxygenated and deoxygenated blood.

(A) Heart (B) Kidney
(C) Liver (D) Anterior pituitary
(E) Posterior pituitary

Options :-

(1) A, B & C

(2) A, D & C

(3) A, E & C

(4) D & C only

70) Which of the following show the correct order of structures involved in generation and propogation of an action potential in the human heart ?

(A) Atrio - ventricular bundle
(B) Atrio - ventricular node
(C) Sino - atrial node
(D) Purkinje fibres
(E) Right and left bundle branches

Choose the correct sequence.

(1) C → B → A → E → D

(2) B → C → A → D → E

(3) C → B → E → A → D

(4) C → A → B → E → D

71) Match the following blood group with their associated Antigens, Antibodies and Donor's group and mark the correct option:-

Blood Group	Antigen on RBCs	Antibodies in Plasma	Donor's Group
(I) A	(i) nil	(A) nil	(a) O
(II) B	(ii) A, B	(B) Anti-A	(b) AB, B, A, O
(III) AB	(iii) B	(C) Anti-B	(c) A,O
(IV) O	(iv) A	(D) Anti-A, Anti B	(d) B,O

Options :-

(1) I-(iv),(C),(c); II-(iii),(B),(d); III-(ii),(A),(b); IV-(i),(D),(a)

(2) I-(i),(A),(a); II-(ii),(B),(b); III-(iii),(C),(c); IV-(iv),(D),(d)

(3) I-(iv),(D),(d); II-(iii),(C),(c); III-(ii),(B),(b); IV-(i),(A),(a)

(4) I-(iii),(C),(d); II-(ii),(B),(c); III-(i),(D),(a); IV-(iv),(A),(b)

72) Which of the following is auto excitable ?

(1) SA node

(2) AV node

(3) Purkinje fibers

(4) All of these

73) If a mother is having blood group 'A'. Rh incompatibility problem occur when the mother and fetus is respectively -

(1) A^+, A^+

(2) A^+, A^-

(3) A^-, A^+

(4) A^-, A^-

74) What is normal systolic blood pressure ?

(1) 120/80

(2) 120

(3) 80

(4) 140/90

75) **Statement-I :-** Contraction of diaphragm increases volume of thoracic chamber in antero-posterior axis.
Statement-II :- An increase in pulmonary volume decreases the intrapulmonary pressure to less than atmospheric pressure for inspiration.

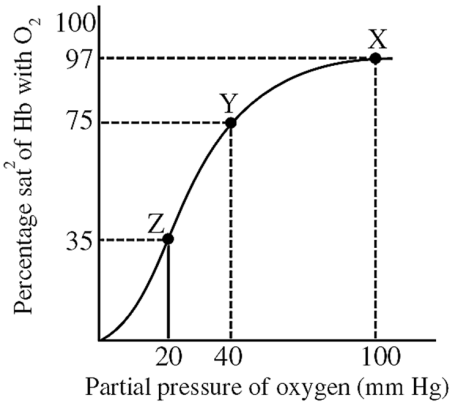
(1) Both statements are correct

(2) Both statements are incorrect

(3) Statement-I is correct and statement-II is incorrect

(4) Statement-I is incorrect and statement-II is correct

76) Given below graph shows an oxygen dissociation curve :-



Identify the correct location of haemoglobin in body with reference to point X, Y and Z of dissociation curve?

(1) X - oxygenated blood in systemic arteries

(2) Y - oxygenated blood in systemic veins (at rest)

(3) Z - oxygenated blood (contracting skeletal muscles)

(4) X - oxygenated blood in pulmonary artery

77) Match the column-I and II and choose the correct option :

	Column-I		**Column-II**
a.	IC	i.	TV + IRV + ERV
b.	EC	ii.	ERV + RV
c.	TLC	iii.	TV + IRV
d.	VC	iv.	FRC + IC
e.	FRC	v.	TV + ERV

(1) a-iii, b-ii, c-iv, d-v, e-i

(2) a-ii, b-i, c-iii, d-v, e-iv

(3) a-iii, b-v, c-iv, d-i, e-ii

(4) a-iii, b-v, c-i, d-iv, e-ii

78) **Assertion :** Human beings have a significant ability to maintain and moderate the respiratory

rhythm to suit the demands of the body tissues.
Reason : A specialised centre present in the medulla region of the brain called respiratory rhythm centre is primarily responsible for regulation of respiration.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) **Assertion** is false but **Reason** is true.

79) Select the sequence of events in inspiration :-
(a) Contraction of diaphragm and external inter-costal muscles.
(b) Thoracic volume increases
(c) Pulmonary volume increases
(d) Pulmonary pressure decreases
(e) Air enters into lungs.
Choose the correct answer from the option given below :-

(1) a, b, c, d, e

(2) b, c, a , d, e

(3) a, c, b, d, e

(4) e, a, b, c, d

80) Emphysema is the chornic respiratory disorder caused mainly by :-

(1) Cigarette smoking

(2) Cheweing tabacco

(3) Inflammation in bronchi and bronchioles

(4) Inflammation in nasal passage

81) Volumn of air remaining in the lungs even after a forcible expiration, this averages _____.

(1) 1000 ml to 1100 ml

(2) 2500 ml to 3000 ml

(3) 1100 ml to 1200 ml

(4) 1200 ml to 1300 ml

82) The maximum volume of air a person can breathe out after a forced inspiration is called :-

(1) Expiratory capacity

(2) Vital capacity

(3) Inspiratory capacity

(4) Total lung caapcity

83) (A) 1100-1200 ml of blood filtered by the kidneys per minute.
(B) A fall in GFR can activate the JG-cells to release renin.
(C) The ascending limb is permeable to water in Henle's loop.
(D) ADH facilitate water reabsorption from later parts of renal tubule (DCT).

(E) Henle's loop and vasa recta play role in counter current mechanism.

(1) A, B, C, & D are correct.
(2) A, B, D, & E are correct.
(3) A, C & E are correct.
(4) A, D & E are incorrect.

84) **Assertion (A) :** Tubular secretion is an important step in urine formation. **Reason (R) :** During urine formation tubular cells secrete like H^+, K^+ and a ammonia to maintain ionic and acid base balance of body fluids.

(1) Both (A) & (R) are true and (R) is not correct explanation of (A)
(2) (A) is true but (R) is false
(3) (A) is false but (R) is true
(4) Both (A) & (R) are true and (R) is correct explanation of (A)

85) Match the items in column-I with those in column-II

	Column-I		**Column-II**
(a)	Uric acid	(i)	Earthworms
(b)	Green glands	(ii)	Crustaceans
(c)	Flame cells	(iii)	Cephalochordate
(d)	Nepheridia	(iv)	Land snails

(1) a-(iv), b-(ii), c-(iii), d-(i)
(2) a-(iv), b-(iii), c-(ii), d-(i)
(3) a-(i), b-(ii), c-(iii), d-(iv)
(4) a-(i), b-(iii), c-(ii), d-(iv)

86) Choose the incorrect statement regarding to aldosterone ?

(1) Aldosterone is released by adrenal cortex.
(2) Maintenance/regulation of water and electrolyte balance in our body.
(3) Help in water reabsorption making the urine hypotonic.
(4) Aldosterone works on the distal part of nephron.

87) Oxalates formed within the kidney :-

(1) Renal calculi
(2) Uremia
(3) Hemodialysis
(4) Glycosuria

88) Glomerular filtration rate (GFR) in a healthy individual is approximately -

(1) 125 litres per day

(2) 180 ml / minute

(3) 180 litre per day

(4) 125 ml / hour

89) Read column-I II & III and identify their correct match :-

Column-I		**Column-II**		**Column-III**	
(A)	Renin	P.	Vasoconstrictor	(i)	Vasodilation
(B)	ADH	Q.	Heart	(ii)	Increases GFR.
(C)	ANF	R.	Posterior pituitary	(iii)	Converts angiotensinogen into angiotensin I
(D)	Angiotensin II	S.	JG cells	(iv)	Reabsorption of water

(1) A-P-i, B-Q-ii, C-R-iii, D-S-iv

(2) A-Q-i, B-P-ii, C-S-iii, D-R-iv

(3) A-Q-i, B-P-iv, C-S-iii, D-R-ii

(4) A-S-iii, B-R-iv, C-Q-i, D-P-ii

90) Secretion of which of the following contains lactic acid:

(1) Lungs

(2) Liver

(3) Sebaceous gland

(4) Sweat gland

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	2	1	1	4	3	1	1	2	4	2	3	3	4	2	1	4	3	1	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	1	1	2	2	2	1	1	1	1	2	2	4	2	1	4	2	2	4	3
Q.	41	42	43	44	45															
A.	1	4	1	2	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	2	2	1	3	1	4	4	1	2	2	4	4	3	1	1	2	3	2	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	2	3	1	3	4	2	2	3	4	1	4	3	3	1	3	3	3	4	4
Q.	86	87	88	89	90															
A.	3	1	4	4	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	1	3	4	2	2	4	3	1	3	1	4	3	1	1	4	1	2	2	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	1	3	2	1	3	2	2	3	3	4	2	3	2	2	2	3	1	2	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	1	1	3	4	2	4	3	3	1	2	3	4	1	3	3	3	2	2	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	3	3	2	2	1	3	3	2	1	1	4	3	2	1	1	3	1	1	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	2	2	4	1	3	1	3	4	4										

SOLUTIONS

PHYSICS

1) $(y_{COM})_i = \frac{a}{2}$

Volume = constant

$(a)^3 = 8(r)^3$

$\Rightarrow r = \frac{a}{2}$

$(y_{COM})_f = \frac{8 \times \frac{a}{2}}{2} = 2a$

Vertical shift = $2a - \frac{a}{2} = \frac{3a}{2}$

2)

Velocity on hitting = $\sqrt{2 \times 10 \times 5} = 10$ m/s

Velocity after first bounce v = $\frac{3}{4} \times 10$ m/s

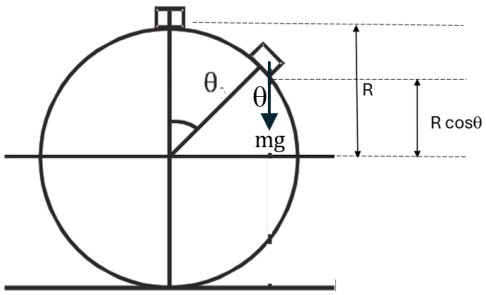
Time = $\frac{2v}{g} = \frac{2 \times 3}{4} \times 10 \times \frac{1}{10} = 1.5$ s

3)

The radius of the sphere be

$R = \frac{30}{2} = 15m$

At angle θ from the top, the particle leaves the surface when normal reaction becomes zero.



Radial force balance.

$mg\cos\theta = \frac{mv^2}{R}$ ------- (i)

Form energy conservation between top and angle θ:

$mgR(1-\cos\theta) = \frac{1}{2}mv^2$

$2mg(1-\cos\theta) = \frac{mv^2}{R}$ ------- (ii)

From eq. (i) and (ii)

$mg\cos\theta = 2mg - 2mg\cos\theta$

$3\,mg\cos\theta = 2\,mg$

$\cos\theta = \frac{2}{3}$

Height from bottom:

$h = R(1+\cos\theta) = 15\left(1+\frac{2}{3}\right) = 25m$

4) Tangential acceleration $a_t = \frac{dv}{dt} = \frac{vdv}{d\ell}$

Solution:

$a_t = \frac{vdv}{d\ell}$

Given :

$v^2 = 4\ell$

$\frac{2vdv}{d\ell} = 4 \Rightarrow \frac{vdv}{d\ell} = 2$

$a_t = 2m/s^2$

5)

Reversing direction changes velocity direction but not the required centripetal force direction (always inward) or magnitude. So magnitude unchanged and direction unchanged. Statements a and c are correct.

6) Minimum coefficient of friction required for pure rolling

$$\mu_{min} = \frac{\tan\theta}{1+\frac{1}{\frac{K^2}{R^2}}} = \frac{\tan 60}{1+\frac{1}{2/5}} = \mu_{min} = 0.495$$

But existing $\mu < \mu_{min}$

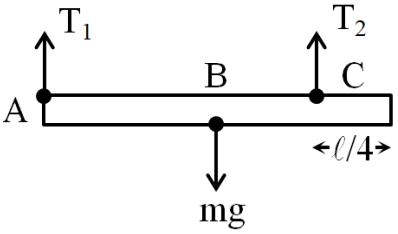
⇒ No pure rolling

7) $I_{remaining} = I_{complete} - I_{removed}$

$$= \frac{9MR^2}{2} - \left[\frac{M(R/3)^2}{2} + M\left(\frac{2R}{3}\right)^2\right]$$

$= 4\,MR^2$

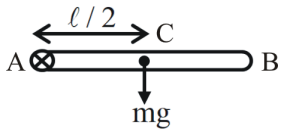
8)



Sum of forces:

$T_1 + T_2 = mg$

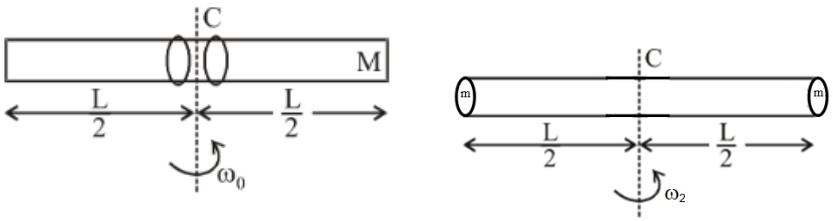
Torque balance about the pivot at A:
$T_2 \times (3\ell/4) = (mg) \times (\ell/2)$
$T_2 = 2mg/3$
BY $T_1 + T_2 = mg$
$T_1 = mg/3$



9) *
$\tau_A = I_A\alpha$
$\Rightarrow mg \times \frac{\ell}{2} = \frac{m\ell^2}{3} \times \alpha$
$\Rightarrow \alpha = \frac{3g}{2\ell}$ (same for all points)
* Acceleration of C : $a_C = \alpha r = \frac{3g}{2\ell} \times \frac{\ell}{2} = \frac{3g}{4}$
Acceleration of B : $a_B = \alpha r = \frac{3g}{2\ell} \times \ell = \frac{3g}{2}$

10) $\theta = \left(\frac{\omega_0 + \omega}{2}\right) t$
$(2\pi)\, n = \left(\frac{10\pi + 30\pi}{2}\right) 10$
$(2\pi)n = 20\pi \times 10$
n = 100 rotations

11)



Suppose, when the beads reach the ends, moment of inertia of the system changes from I_1 to I_2 and therefore, the angular velocity changes from ω_1 to ω_2. As external torque acting on the system is zero, hence angular momentum of the system is conserved.

i.e. $I_1\omega_1 = I_2\omega_2$

$$\left(\frac{ML^2}{12} + 0\right)\omega_0 = \left[\frac{ML^2}{12} + 2m\left(\frac{L}{2}\right)^2\right]\omega_2$$

$$\omega_2 = \left[\frac{\frac{ML^2}{12}}{\frac{ML^2}{12} + \frac{mL^2}{2}}\right]\omega_0$$

or $$\omega_2 = \left(\frac{M}{M + 6m}\right)\omega_0$$

12) $V = -\frac{GM}{R} + \left(-\frac{Gm}{x}\right) = -G\left[\frac{M}{R} + \frac{m}{x}\right]$

13) $g' = g - R\omega^2\cos^2 60 = 0$

$\omega^2 = \frac{4g}{R}$ or $\omega^2 = 2\sqrt{\frac{g}{R}}$

$\frac{2\pi}{T} = 2\sqrt{\frac{g}{R}}$

$T = \pi\sqrt{\frac{R}{g}}$

14) $L = mvr = m\sqrt{\frac{GM}{r}}\ r = m\sqrt{GMr}$

$L \propto \sqrt{r}$

$\frac{L_1}{L_2} = \sqrt{\frac{r_1}{r_2}} = \sqrt{\frac{r}{16r}}$

$L_2 = 4L_1 = 4L$

15) Velocity required to escape : $v_e = \sqrt{\frac{2GM}{R}}$

Velocity required to orbit : $v_o = \sqrt{\frac{GM}{R}}$

$\frac{v_e}{v_o} = \sqrt{2}$

16) Gravitational field intensity at point O is zero (as cavities are symmetrical with respect to O)

17)

Height of liquid rise in a capillary tube: $H = \frac{2T\cos\theta}{r\rho g}$

Potential energy of the water column: $U = mgh_{cm}$

When water rise upto a height h then mass of liquid rise $m = (\pi r^2 h)\rho$

- total mass be located at centre of mass.

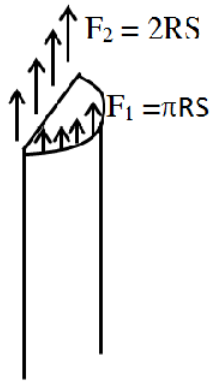
then potential energy $U = mg\frac{h}{2} = (\pi r^2 h\rho)g\frac{h}{2}$

by Zurin law $r \times h$ = constant

- U = constant

18) $\pi R \times S = 2\ RS$

$\pi : 2$



19) Let in water immersed volume is V_1 then

$w = (Th)_o + (Th)_w$

$V\rho_B g = (V-V_1)\rho_o g + V_1 \rho_w g$

$V \times 0.9 \times 10^3 = (V-V_1) \times 0.8 \times 10^3 + V_1 \times 10^3$

$V(0.9\times10^3 - 0.8\times10^3) = V_1 (10^3 - 0.8\times10^3)$

$$\frac{V_1}{V} = \frac{1}{2}$$

20) $$\frac{t_1}{t_2} = \frac{\sqrt{H} - \sqrt{\frac{H}{4}}}{\sqrt{\frac{H}{4}} - 0} = 1$$

21) $$\Delta P = \frac{1}{2}\rho(v_2^2 - v_1^2)$$

$F = A\Delta P$

22) Let the original length of the string be L.

Applying $F = kx$

we have $4 = k(5 - L)$

$5 = k(6 - L)$

$9 = k(2X - L)$

From these equations, X= 5

23) $$\frac{1}{2}Y \text{ (strain)}^2 \text{ (volume)} = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{Y \times \text{volume}}{m}} \times \text{strain}$$

$$v = \sqrt{\frac{6\times10^8}{10^{-2}}\left(\frac{1}{6}\times10^{-4}\right)} \times \left(\frac{6}{1}\times\frac{1}{6}\right)\times10^{-2}$$

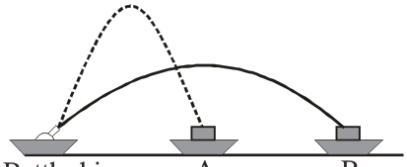
$v = 10^3 \times 10^{-2}$

$v = 10 m/s$

24)

$F = \eta A \frac{dv}{dz}$

25) $v = t - t^3$ average velocity $= \int_0^2 \frac{(t - t^3)\,dt}{2} = -1$



26)

$H = \frac{u^2 \sin^2\theta}{2g}$

$T = \frac{2u \sin\theta}{g}$

As maximum height for A is more than B, hence its time of flight will also be large compared to time of flight for ship B.

$T_A > T_B$

Hence, ship B will get hit first.

27) $\frac{dV}{dt} = -a\sqrt{v}$

$\frac{dV}{v^{1/2}} = -adt$

$\int v^{-1/2} dv = \int -adt$

$2\sqrt{v} = -at + c$

at $t = 0$ $v = V_0$

$\Rightarrow 2\sqrt{v} = -at + 2\sqrt{V_0}$

put $v = 0$

$t = \frac{2\sqrt{V_0}}{a}$

28) $24.5 = \frac{g}{2}(2n - 1) \Rightarrow n = 3$

$h = \frac{1}{2} g n^2 = \frac{1}{2} \times 9.8 \times 9 = 44.1\,m$

29) When two particles moves towards each other then

$v_1 + v_2 = 6$ ....(i)

When these particles moves in the same direction then

$v_1 - v_2 = 4$ ....(ii)

By solving $v_1 = 5$ and $v_2 = 1$ m/s

30) $\vec{r_f} = \vec{r_i} + \vec{u}t + \frac{1}{2}\vec{a}t^2$

$\vec{r_f} = 2\hat{i} + 4\hat{j} + 10\hat{i} + 8\hat{j} + 8\hat{i} + 8\hat{j}$
$\vec{r_f} = 2\hat{i} + 4\hat{j} + 18\hat{i} + 16\hat{j}$
$\vec{r_f} = 20\hat{i} + 20\hat{j}$
distance $= 20\sqrt{2}$m

31)

For I ball : $h = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times 18^2 = 1620m$
For II ball : (h = 1620m, u = v, t = 18 – 6 = 12s)
$h = ut + \frac{1}{2}gt^2$; $1620 = 12v + \frac{1}{2} \times 10 \times 12^2$
1620 = 12v + 720
12v = 900 ⇒ v = 75 m/s

32) S = emf per unit temperature difference
σ = Electrical conductivity
k = Thermal conductivity
$[S] = [ML^2T^{-3}I^{-1}K^{-1}]$
$[\sigma] = [M^{-1}L^{-3}T^3I^2]$
$[K] = [M^1L^1T^{-3}K^{-1}]$
$[Z] = \frac{S^2\sigma}{K} = \frac{\left[M^1L^1T^{-3}K^{-2}\right]}{\left[M^1L^1T^{-3}K^{-1}\right]}$
$[Z] = [K^{-1}]$

33) (A) Trailing zero is non-significant
$12 \cdot 5849$
(B) $\frac{2 \cdot 4}{14.9849}$
= 15.0
(C) Trailing zero after decimal is significant figure.

34) Leat count = $\frac{0.5mm}{100}$
Positive error = $2 \times \frac{0.5mm}{100} = \frac{1}{100}mm$
Scale reading = 8 × 0.5 mm
$+ 83 \times \frac{0.5mm}{100} - \frac{1}{100}mm$
⇒ 4.405 mm

35) $W = \int Fdx$
$= \int_0^1 (ax - bx^2)dx$

$$= \left[\frac{ax^2}{2} - \frac{bx^3}{3}\right]_0^1$$

$$= \frac{a}{2} - \frac{b}{3}$$

$$= \frac{1}{2} - \frac{1}{3} \qquad (\because a = 1\ N/m,\ b = 1\ N/m^2)$$

$$= \frac{1}{6}J$$

36) $U = \frac{1}{2}K(1)^2$ ...(1)

$$U' = \frac{1}{2}K[(4)^2 - (1)^2]$$

$$U' = \frac{1}{2}K[15]$$ ...(2)

From (2) ÷ (1)

$$\frac{U'}{U} = 15$$

$$\boxed{U' = 15U}$$

37) $F = -\frac{dU}{dr}$

$$F = \frac{2B}{r^3} - \frac{3A}{r^4}$$

For equilibrium

$F = 0$

$$\frac{2B}{r^3} = \frac{3A}{r^4}$$

$$r = \frac{3A}{2B}$$

38)

$P = FV$

$P = ma\nu$ $\nu = u + at = at$

$= (ma)at$

$P = ma^2t$

$P \propto t$

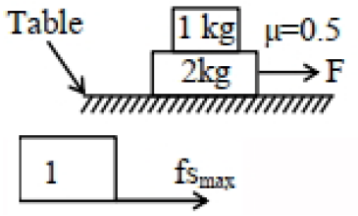
39) $F = 3a$ (For system) ....(i)

$fs_{max} = 1a$ (for 1 kg block) ....(ii)

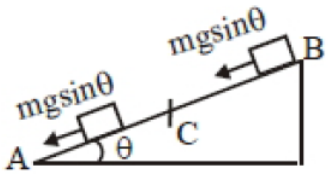
$\mu \times 1 \times g = 1\ (a)$

$\Rightarrow 5 = a$

$F = 15N$



40) Apply work energy theorem
mg sinθ (AC + 2AC) – μmg cosθ AC = 0
μ = 3 tanθ



41) Acceleration of system of blocks = $\frac{F}{8+2} = \frac{F}{10} ms^{-2}$
Now, from free body diagram below, we see that
Reaction of X and Y is

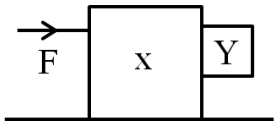
$R = m_Y a = 2 \times \frac{F}{10}$

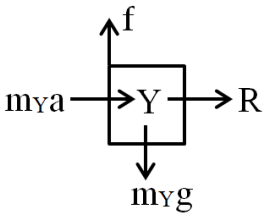
$\Rightarrow R = \frac{F}{5}$

Block Y does not slips, if
Force of friction = Weight of Y

$\Rightarrow \mu R = m_Y g \Rightarrow 0.5 \times \frac{F}{5} = 2 \times 10$

So, F = 200 N





42) $F_A = \frac{\mu mg}{\cos 30^\circ - \mu \sin 30^\circ}$

$F_B = \frac{\mu mg}{\cos 60^\circ + \mu \sin 60^\circ}$

$\frac{F_A}{F_B} = \frac{\cos 60^\circ + \mu \sin 60^\circ}{\cos 30^\circ - \mu \sin 30^\circ}$

$\frac{F_A}{F_B} = \frac{2}{\sqrt{3}}$

43) Case I : In downward movement the body is in equilibrium

f_V = (wt of body)effective
Case II: In upward movement
downward force = $F_V + E_{ff} \cdot$ weight
⇒ Downward force = 2 ($E_{ff} \cdot$ weight)
⇒ Downward force = upward force
So ball is still in equilibrium and move with same speed 10 cm/s.

44) **Solution :** 20 VSD = 19 MSD

$$1\,\text{VSD} = \frac{19}{20}\text{MSD}$$

L.C. = 1 MSD – 1 VSD

$$0.2\,\text{mm} = 1\,\text{MSD} - \frac{19}{20}\text{MSD}$$

0.2 × 20 mm = 1 MSD

$$\boxed{\text{MSD} = 4\,\text{mm}}$$

45)

Solution :
$F - F_k = 4a$

$60 - F_k = 40$
⇒ $F_k = 20$
$\mu_k (4g) = 20$
$\mu_k = 0.5$

CHEMISTRY

46) **Solution:** Small 2p orbitals overlap effectively forming strong π bonds; large 3p orbitals overlap
weakly, reducing stability of S_2 and P_2. The reason correctly explains the assertion.
Final Answer: 1

47) **Solution:**
I_3^-: Linear → 180°
BCl_3: Trigonal planar → 120°
PCl_5: Trigonal bipyramidal → 120° (equatorial)
CO_3^{2-}: Trigonal planar → 120°
CH_3^-: Trigonal Pyramidal → ≈107°
SF_6: Octahedral → 90°, 180°
XeF_2: Linear → 180°
SO_3: Trigonal planar → 120°
Total = 4 species (BCl_3, PCl_5, CO_3^{2-}, SO_3)
Final Answer: 2

48)

$XeOF_2$ → T-Shape
PCl_5 → Trigonal Bi Pyramidal
IF_5 → Square pyramidal
CCl_4 → Tetrahedral
XeF_4 → Square planar

49) **Solution :** In PCl_5 structure (trigonal bipyramidal) there are 2 axial and 3 equatorial bonds so bond lengths are different as Bond length (axial > equatorial) Because each of the two axial P-Cl bonds is at a 90° angle to the three equatorial P-Cl bonds. This close proximity results in significant bond pair-bond pair repulsion. Therefore both assertion & reason are true & the reason is a correct explanation of the assertion.
Final answer : Option(1)

50) By factors

51)

Concept

52)

Boiling point order → BiH_3 >SbH_3 > NH_3 > AsH_3 > PH_3

53) **Solution :** Statement 4 is wrong. N_2^- has an unpaired electron in an antibonding orbital, making it paramagnetic, not diamagnetic.
N_2^-(15 electrons) - $\sigma 1s^2$, $\sigma^* 1s^2$, $\sigma 2s^2$, $\sigma^* 2s^2$, $\pi 2px^2 = \pi 2py^2$, $\sigma 2pz^2$, $\pi^* 2px^1$
Final Answer : option (4)

54)

Explaining the Question :- We are given ionic potential order : $M_a^+ > M_b^+ > M_c^+$ and polarizability order: $x^- > y^- > z^-$.
We must select the ion pair with maximum covalent character.
Solution :-
• High ionic potential (small, highly charged cation) → more covalent.
• High polarizability (large, easily distorted anion) → more covalent.
• So, $M_a^+X^-$ has both highest cationic potential and highest anionic polarizability → maximum covalent character.
Final Answer :- 1

55) **Solution:**
Formal charge (FC) is a hypothetical charge assigned to atoms in a Lewis structure.
The most stable structure usually has minimum formal charges on atoms and a proper distribution of charges. FC helps compare possible structures and pick the lowest energy one.
Statement-I: Correct, because FC indeed helps in selecting the lowest energy structure from many possibilities.

Statement-II: Correct, because generally the most stable structure has the smallest (closest to zero) formal charges on atoms.
Both Statement-I and Statement-II are correct.
Final Answer: (2)

56) [Rn] $7s^2\ 6d^2$ – d-block
[Xe] $6s^2\ 4f^{14}\ 5d^1$ – f-block
[Xe] $6s^2\ 4f^{14}\ 5d^{10}\ 6p^1$ – p-block
[Xe] $6s^2\ 4f^1\ 5d^1$ – f block

57)

Size ∝ Shell no. (n)

58)

Solution : After losing one electron, element B has a single p-electron, which is easiest to remove (lowest IE_2). And After losing one electron, Element D has a stable half-filled p-orbital (D^+ is ns^2np^3), making it hardest to remove from (highest IE_2).
Final answer : 4

59) NCERT XIth Page no. 87, table 3.7

60) **Solution:**
• **Statement I is true:** The acid strength indeed increases in the order HF<< HCl << HBr << HI.
• **Statement II is also true:** As we move down the group, the size of the halogen atom increases. This leads to a weaker H-X bond, making it easier to release the proton. Hence, the acid strength increases.
Therefore, the correct answer is **option 1: Both Statement I and Statement II are true.**

61)

Solution :-
CO : Neutral oxide
BaO : Basic oxide
Al_2O_3 : Amphoteric oxide
Cl_2O_7 : Acidic oxide
Final answer :- (1)

62)

Recently discovered element of group '14' is Fl (at. no. 114)
Ununquadium

63) Theory based.

64) From (n + ℓ) Rule

65)

Order of stability (higher K → more stable)
NO_2 (8×10^{-31}) > NO (5×10^{-31}) >> N_2O_5 (10^{-34})

66) A ⇌ B; $K_1 = 100$ ...(1)
3C ⇌ 3B; $K_2 = 10^6$
3B ⇌ 3C, $K_2' = 10^{-6}$
B ⇌ C; $K_2'' = 10^{-2}$ ...(2)
½C ⇌ ½D; $K_3 = 10^{-4}$
C ⇌ D; $K_3'' = 10^{-8}$ ...(3)
Add eq^n, 1, 2 and 3
A ⇌ D; $K_1 \cdot K_2'' \cdot K_3''$
$= 10^2 \times 10^{-2} \times 10^{-8} = 10^{-8}$

67) **Solution :**

$PCl_5(g) \rightarrow PCl_3(g) + Cl_2(g)$

moles at eq $1-\alpha$ α α

$$K_P = \frac{\left(P_{PCl_3}\right)\left(P_{Cl_2}\right)}{P_{PCl_5}} = \frac{\left(\frac{\alpha P}{1+\alpha}\right)\left(\frac{\alpha P}{1+\alpha}\right)}{\left(\frac{1-\alpha}{1+\alpha}\right)P} = \frac{\alpha^2 P}{1-\alpha^2}$$

Since K_p, is constant, $\alpha^2 \propto \frac{1}{P}$, which leads to $\alpha \propto \frac{1}{\sqrt{P}}$

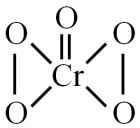
68) $E_{Br_2} = \frac{M}{2} + \frac{M}{10} = \frac{3M}{5}$

69)

S_8 (ON = 0)
$S_2O_3^{2-}$ (ON of S = +2)
$S_2O_7^{2-}$ (ON of S = +6)
$S_4O_6^{2-}$ (ON. of S = +2.5)
SO_3^{2-} (ON. of S = +4)

70)

CrO_5 →



71) $(\text{mole} \times \text{v.f.})_{KMnO_4} = (\text{mole} \times \text{v.f.})_{SnC_2O_4}$

72) $$\frac{\lambda_{2\to1}}{\lambda_{\infty\to2}} = \frac{\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right)}{\left(\frac{1}{1^2} - \frac{1}{2^2}\right)} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3} = \frac{x}{\text{unknown}}$$

Unknown = 3x

73)

Solution:

First, calculate the uncertainty in velocity (Δv) from the given percentage.

Then, use the minimum uncertainty relation to find Δx.

Given: Mass of electron (m) $=9.1\times10^{-31}$ kg

Velocity (v) $=300$ m s^{-1}

Percentage uncertainty in velocity

$\left(\frac{\Delta v}{v} \times 100\right) = 0.001\%$

Planck's constant (h) $= 6.63 \times 10^{-34}$ J s

A. Calculate Δv: $\Delta v = v \times \frac{0.001}{100} = 300 \times 10^{-5} = 3 \times 10^{-3}$ m s^{-1}
B. Calculate Δx using the minimum uncertainty relation:

$$\Delta x = \frac{h}{4\pi m \Delta v}$$

$$\Delta x = \frac{6.63 \times 10^{-34}}{4 \times 3.14159 \times 9.1 \times 10^{-31} \times 3 \times 10^{-3}}$$

$$\Delta x = \frac{6.63 \times 10^{-34}}{34.3 \times 10^{-33}} \approx 0.193 \times 10^{-1} \text{ m}$$
$$= 1.93 \times 10^{-2} \text{ m}$$

74)

$$E = \frac{nhc}{\lambda}$$

$$P \times t \times \frac{50}{100} = \frac{nhc}{\lambda}$$

$$360 \times 1 \times \frac{50}{100} = \frac{n \times 6.62 \times 10^{-34} \times 3 \times 10^{8}}{331 \times 10^{-9}}$$

$n = 3 \times 10^{20}$

$x = 3$

75) Mass of $CO_2 = \frac{1120}{22400} \times 44 = 2.2$ g

76)

moles of $SO_2 = \frac{11.2}{22.4} = 0.5$

	SO_2	+ $2H_2S$	→ $3S$	+ $2H_2O$
n_i	0.5	2	0	0
	(LR)			
n_f	0	1	1.5	1

77) $n_{Fe_2O_3} = \frac{100 \times 80}{100} \times \frac{1}{160}$

$w_{Fe} = \frac{100 \times 80}{100 \times 160} \times 2 \times 56$

78)

Mathematical Calculation:
Let M be the minimum molecular weight of the compound. We can set up the equation:
(Mass of sulphur/M) × 100 = Percentage of sulphur
(32/M) × 100 = 4
3200 = 4M
M = 3200/4
M = 800

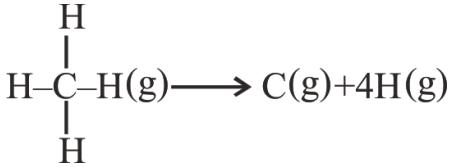
79)

Adiabatic free expansion for ideal gas
q = 0, w = 0, ΔE = 0, ΔT = 0

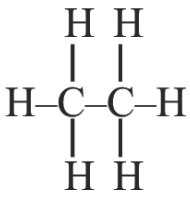
80)

$\Delta G_{sys} = -T\Delta S_{Total}$
$\Delta G_{sys} = \Delta H_{sys} - T\Delta S_{sys}$

81) ΔG = ΔH – TΔS
For spontaneous process, ΔG is –ve i.e., < 0
∴ ΔH – TΔS < 0 or TΔS > ΔH
 TΔS and ΔH are +ve
Thus ΔG = ΔH – TΔS become –ve.

82) 

⇒ BE of C–H bond = $\frac{360}{4}$ kJ/mol


has 6C–H bond and one C–C bond

$\Delta H_{C-C} + 6\Delta H_{C-H} = 620$ kJ/mol

$\Delta H_{C-C} = 620 - 6 \times 90$

$= 80$ kJ/mol

83) $CH_4 + 2CO_2 \rightarrow CO_2 + 2H_2O$

 For 2 mole or 36 gm water,

heat released = –400 kJ/mol

 For 40 gm water,

heat released will be = $\frac{-400}{36} \times 40 = -444.44$ kJ

84) $T > \frac{\Delta H}{\Delta S}$

85)

$Ag_3PO_4 \rightleftharpoons 3Ag^+ + PO_4^{-3}$

$(3s + 0.1) \quad s$

$Ksp = (3s + 0.1)^3(s) = 1.1 \times 10^{-16}$

$10^{-3} \times s = 1.1 \times 10^{-16}$

$s = 1.1 \times 10^{-13}$

86)

Intensive properties → Mass and size independent

Temperature, density and pressure are intensive properties.

87) NaOCN is a salt of a strong base (NaOH) and a weak acid (HOCN). The OCN^- ion undergoes hydrolysis to produce OH^- ions. The concentration of OH^- can be calculated using the formula for weak base equilibrium.

Given: C = 0.01 M, $K_b = 10^{-10}$.

$[OH^-] = \sqrt{K_b \times C}$

$[OH^-] = \sqrt{10^{-10} \times 0.01} = \sqrt{10^{-12}}$

$[OH^-] = 10^{-6}M$

88) Final $[H^+] = \frac{10^{-3} + 10^{-6}}{2} = 5 \times 10^{-4}$

$-\log [H^+] = pH = 3.3$

89)

By reducing the total pressure or increasing the total volume, the reaction will shift to the side having more no. of moles.

90)

Concept : This question is based on
1. $\Delta H = \Delta U + \Delta n_g RT$
2. $\Delta G = \Delta H - T\Delta S$

Solution :
$\Delta ng = 2 \text{ mol} - 0 \text{ mol} = 2 \text{ mol}$
$\Delta H = 2100 \text{ cal} + (2 \times 2 \times 300) \text{ cal}$
$\Delta H = 2100 \text{ cal} + 1200 \text{ cal} = 3300 \text{ cal} = 3.3 \text{ kcal}$
$\Delta G = 3.3 \text{ kcal} - (300 \text{ K} \times 0.020 \text{ kcal/K})$
$\Delta G = 3.3 \text{ kcal} - 6.0 \text{ kcal} = -2.7 \text{ kcal}$

BIOLOGY

91) NCERT Pg#129 (8.5) Pg#126 (last line) Pg#127 (last line of 1st para) Pg#134 (8.5.4) Pg#138 (8.5.10)

92) NCERT-XI, Pg. # 98

93) NCERT Pg. No. 98, 99, 102

94) NCERT (XIth) Pg. # 129

95)

NCERT Pg#127

96) NCERT XI Pg. # 138, 140

97)

NCERT-XI, Pg. # 93, 94

98)

NCERT-XI, Pg. # 94

99)

NCERT-XI, Pg. # 121, 122

100) NCERT-XI, Pg.# 126

101)

NCERT-XI, Pg. # 121-124

102)

NCERT XI, Page # 134

103)

NCERT-XI, Pg. # 127, 128

104)

NCERT-XI, Pg. # 126, 127

105)

NCERT-XII, Pg # 167,168,169

106)

NCERT-XI, Pg. # 106

107)

NCERT-XI, Pg. # 117

108)

NCERT-XI, Pg. # 112

109)

NCERT XI, Page # 109

110)

NCERT-XI, Pg. # 112, 113

111)

NCERT-XI, Pg. # 108-110

112)

Module, chapter Enzyme

113)

NCERT-XI, Pg. # 6,7, 8

114)

Old NCERT-XI, Pg. # 5
The correct answer is:
A-Incorrect; B - Correct

115) NCERT, Pg. # 7

116) NCERT Pg. No. # 7, 8
Statement c and d are incorrect

117) NCERT PG. 10, 11

118)

NCERT Pg. # 17

119)

NCERT-XI, Pg. # 13

120)

NCERT-XI, Pg. # 13

121) NCERT XI, Pg # 14

122)

NCERT-XI, Pg. # 13

123)

NCERT-XI, Pg. # 15

124)

NCERT-XI, Pg. # 16

125)

NCERT-XI, Pg. # 12,15,17,26

126)

NCERT XI, Pg # 22

127)

NCERT-XI, Pg. # 24

128) NCERT pg.no. (E), (H) 34

129) NCERT Pg. # 39

130)

NCERT XI, Pg # 33

131) NCERT XI, Pg.No.# 38

132)

NCERT-XI, Pg. # 32, 33

133) NCERT XI Pg#33,34

134)

NCERT-XI, Pg. # 25, 26, 27

135)

NCERT XI Pg # 23, 24

136)

NCERT-XI, Pg. # 48,49

137) NCERT Pg. # 47, 48

138) NCERT Pg. # 48

139) NCERT Pg. # 48

140) NCERT Pg. # 47

141) NCERT Pg. # 46

142)

NCERT Pg. # 50

143)

NCERT, Pg. # 50

144)

NCERT (XIth) Pg. # 42, 45

145)

NCERT Pg.#44-46.

146)

NCERT Pg. # 51

147) NCERT-XI, Pg. # 101

148) Old NCERT
Module

149) NCERT Pg. # 104

150) Module

151) Module

152)

NCERT_XII_Pg. No. 114

153) NCERT Page - 118 and 119

154) NCERT Page-116

155) NCERT Page No. 118

156) NCERT Page No. 118

157) NCERT Pg. # 104

158) NCERT Class-XII, Page No. 199, 200

159) NCERT Pg. # 201

160) NCERT XI, Pg. # 199

161) NEW NCERT Biology XI Page no. 195

162) NCERT Pg.# 199

163) NCERT Pg. No. # 196

164) NCERT Pg. # 202

165) NCERT Pg. # 185, 186

166) NCERT XI, Pg. # 189, Fig. 14.5

167) NCERT, Pg. # 187

168) NCERT Pg.# 190

169) NCERT Pg. # 185

170) NCERT Pg. # 190

171) NCERT Pg. # 272

172) NCERT Pg. # 186-187

173) NCERT, Pg. # 208, 209, 211, 212

174) NCERT-XI, Pg. # 209

175) NCERT Pg. # 206

176) NCERT_Pg. NO. 212

177) NCERT Pg. # 213, 214

178) NCERT Pg. 209

179) NCERT Pg. # 212

180) NCERT Pg. No. # 213